

04 — Research Design Document

Interpretable Machine Learning for Discovering Hidden Regulatory Switches in Non-Coding DNA

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1. Scientific Background

1.1 The Non-Coding Genome and Gene Regulation

The human genome consists of approximately 3.2 billion base pairs, of which only ~1.5–2% encodes proteins. The remaining ~98% — the non-coding genome — was historically dismissed as "junk DNA," but decades of research have revealed that it contains critical regulatory elements that control gene expression.

Key types of non-coding regulatory elements:

Element	Function	Typical size	Distance from gene
Promoter	Initiates transcription; RNA polymerase binding site	100–1000 bp	0–2 kb upstream of TSS
Enhancer	Increases transcription of target genes; can act over long distances	200–1500 bp	1 kb – 1 Mb from target
Silencer	Represses transcription	200–1000 bp	Variable
Insulator	Blocks enhancer-promoter communication	300–2000 bp	Variable
Open chromatin region	Accessible DNA indicative of regulatory activity	Variable	Variable

These elements function through binding of transcription factors (TFs) to specific DNA sequence motifs, recruitment of co-activators or co-repressors, and modulation of local chromatin structure.

1.2 The Regulatory Switch Concept

In this project, a "regulatory switch" is defined as:

A sequence region or local sequence-pattern combination in non-coding DNA that appears to strongly influence regulatory activity — such as enhancer-like behavior, promoter-like behavior, chromatin accessibility, or context-specific control of gene expression.

This is a **working project definition**, not a claim of molecular proof. The term "switch" reflects the idea that specific sequence patterns can toggle regulatory states (active/inactive), analogous to a switch controlling a circuit. The project aims to discover candidate switch-like patterns computationally.

1.3 Computational Approaches to Regulatory Prediction

Sequence-based regulatory prediction is an established computational direction. Key approaches include:

1. **Motif-based methods:** Scan sequences for known TF binding motifs (e.g., using PWMs from JASPAR). Simple, interpretable, but limited to known motifs.
2. **k-mer-based methods:** Represent sequences as frequency vectors of all subsequences of length k. Captures composition without requiring known motifs.
3. **Convolutional neural networks (CNNs):** Learn sequence features automatically via convolutional filters. DeepSEA, Basset, and similar models demonstrated this approach.
4. **Recurrent / attention models:** Capture long-range dependencies. DanQ combined CNN + RNN for regulatory prediction.
5. **Genomic foundation models:** Large pretrained models (Enformer, DNABERT, Nucleotide Transformer, HyenaDNA) trained on massive genomic corpora. High accuracy but require substantial computational resources for training.

1.4 The Interpretability Gap

Machine learning in genomics involves trade-offs between predictive performance and human interpretability:

- **High interpretability, lower capacity:** Logistic regression on k-mers or motif features. Every prediction can be traced to specific features. May miss complex interactions.
- **Medium interpretability:** Shallow CNNs where learned filters can be mapped to motifs. SHAP or saliency methods provide post-hoc explanations.
- **Low interpretability, higher capacity:** Large transformers. Strong predictions but difficult to explain which sequence elements drive the output.

This project explicitly operates in the high-to-medium interpretability zone, prioritizing biological insight over raw accuracy.

1.5 ENCODE as a Data Foundation

The Encyclopedia of DNA Elements (ENCODE) project aims to identify all functional elements in the human and mouse genomes. It provides:

- Chromatin accessibility maps (DNase-seq, ATAC-seq)
- Histone modification profiles (ChIP-seq for H3K27ac, H3K4me3, H3K4me1, etc.)
- Transcription factor binding sites (ChIP-seq)
- Candidate cis-regulatory elements (cCREs) — pre-classified regulatory annotations

ENCODE's candidate cis-regulatory elements (cCREs) are specifically designed to identify regulatory regions and serve as appropriate labels or candidate regions for this project.

2. Hypothesis Framing

2.1 Central Hypothesis

Non-coding DNA regions annotated as regulatory by ENCODE contain sequence-level patterns (motifs, k-mer compositions, local subsequence arrangements) that are learnable by interpretable machine learning models and that correspond to known or novel regulatory mechanisms.

2.2 Operational Hypotheses

ID	Hypothesis	Testable by
H1	ENCODE-annotated cCREs have distinguishable k-mer composition from randomly sampled non-coding regions.	Train k-mer classifier; AUROC > 0.5
H2	Top-importance k-mers or features overlap with known TF binding motif cores.	Cross-reference top features with JASPAR/HOCOMOCO
H3	Regulatory classification accuracy differs across cell types, reflecting cell-type-specific regulation.	Compare AUROC across K562, GM12878, HepG2 models
H4	Different regulatory element types (enhancer vs. promoter) have distinct sequence signatures.	Train separate classifiers; compare top features
H5	Shallow CNN filters learn motif-like patterns that overlap with known TF motifs.	Visualize CNN filters; compare to JASPAR PWMs
H6	Combining interpretable features with pretrained embeddings improves prediction without destroying interpretability.	Stage 3 experiment

3. Research Questions

3.1 Primary Research Questions

#	Question
RQ1	Can a lightweight, interpretable ML model distinguish ENCODE-annotated regulatory regions from non-regulatory non-coding DNA with AUROC ≥ 0.80 ?
RQ2	Which sequence features (k-mers, motifs, compositional statistics) are most predictive of regulatory activity, and do they align with known biology?
RQ3	How do regulatory sequence signatures differ between cell types (e.g., K562 vs. GM12878)?

3.2 Secondary Research Questions

#	Question
RQ4	Does a shallow CNN discover motif-like convolutional filters, and how do they compare to known TF motifs?
RQ5	What is the interpretability-accuracy trade-off when comparing logistic regression, tree-based models, CNNs, and pretrained-embedding classifiers?

RQ6	Can the pipeline identify novel candidate regulatory patterns not currently in motif databases?
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4. Experimental Design

4.1 Experiment Matrix

Experiment	Independent variable	Dependent variable	Controls
E1: Baseline classification	Model family (LogReg, RF, XGB)	AUROC, AUPRC, F1	Same data split, same features, same seed
E2: Feature ablation	Feature set (k-mer only, k-mer+GC, k-mer+GC+motif)	AUROC delta	Same model (RF), same split
E3: k-mer resolution	Value of k (3, 4, 5, 6)	AUROC, top features	Same model (RF), same split
E4: Cell-type comparison	Cell type (K562, GM12878, HepG2)	AUROC, top features overlap	Same model, same pipeline
E5: Element-type classification	Element type (enhancer-like, promoter-like)	AUROC, top features	Same model, same split
E6: CNN motif discovery	—	Learned filter similarity to JASPAR	Fixed architecture
E7: Negative set sensitivity	Negative sampling strategy	AUROC variation	Same model, same positives
E8: Pretrained embeddings	Feature source (k-mer vs. pretrained)	AUROC, interpretability	Same classifier head

4.2 Experiment Execution Order

E1 (baseline)
■■■→ E3 (k-mer resolution) — informs feature choice for remaining experiments
■■■→ E2 (Feature ablation) — determines final feature set
■■■→ E7 (Negative set sensitivity) — validates data robustness
■■■→ E4 (Cell-type comparison)
■■■→ E5 (Element-type classification)
■■■→ E6 (CNN motif discovery)
■■■→ E8 (Pretrained embeddings) — Stage 3 only

5. Dataset Strategy

Detailed in [Document 05 — Dataset Strategy](#). Summary:

Dataset	Use	Phase
ENCODE cCREs (human, hg38)	Primary labels for regulatory regions	Phase 1
hg38 reference genome	Sequence extraction	Phase 1

JASPAR motif database	Motif cross-referencing	Phase 1–2
ENCODE DNase-seq peaks	Cell-type-specific accessibility labels	Phase 2
ENCODE histone ChIP-seq peaks	Enhancer/promoter mark labels	Phase 2
GTEx expression data	Expression-linked regulatory analysis	Phase 3+

6. Baselines

6.1 Trivial Baselines (Must Outperform)

Baseline	Expected AUROC	Purpose
Random classifier	0.50	Sanity check
GC-content-only logistic regression	0.55–0.65	Tests whether composition alone is predictive
Single-feature (region length)	~0.50	Tests data leakage through length

6.2 Interpretable Baselines (Project Core)

Model	Features	Expected AUROC range
Logistic regression	k-mer (k=4,5,6)	0.70–0.80
Random forest	k-mer + GC + dinucleotide	0.75–0.85
XGBoost	k-mer + GC + dinucleotide + motif	0.78–0.88

6.3 Deep Learning Baselines (Phase 2)

Model	Input	Expected AUROC range
Shallow CNN (2–3 conv layers)	One-hot encoded sequence	0.80–0.90
CNN + SHAP/saliency	One-hot encoded sequence	Same accuracy, +interpretability

6.4 Reference Points (Literature)

Published method	Task	Reported AUROC	Note
DeepSEA (2015)	Multi-task regulatory prediction	~0.90+ per task	Large model, extensive training
Basset (2016)	Chromatin accessibility	~0.89 mean	CNN on one-hot
gkmSVM	Regulatory vs. non-regulatory	~0.85–0.90	Gapped k-mer SVM; interpretable

These are reference points only. The project does not aim to beat state-of-the-art large models but to achieve competitive performance with interpretable methods.

7. Model Comparison Plan

7.1 Standardized Evaluation Protocol

Every model evaluated using:

1. **Same data:** Identical train/val/test split (chromosome-holdout).
2. **Same metrics:** AUROC, AUPRC, accuracy, F1, precision, recall.
3. **Same seed:** Random seed = 42 for all stochastic operations.
4. **Same reporting:** Standardized metrics table + interpretability report.

7.2 Comparison Dimensions

Dimension	How measured
Predictive accuracy	AUROC, AUPRC on test set
Calibration	Calibration plot (predicted probability vs. observed frequency)
Interpretability utility	Qualitative: do top features match known biology?
Interpretability depth	Number of samples for which explanations are available
Compute cost	Training time (seconds), memory usage (MB)
Kaggle feasibility	Does it run within Kaggle constraints? (binary)
Robustness	Performance variance across 5-fold CV on training data

7.3 Output: Model Comparison Matrix

Model	Interpretable?	Kaggle?
LogReg	Yes	Yes
RF	Yes	Yes
XGBoost	Yes (SHAP)	Yes
CNN	Partial	Yes
Pretrained	Limited	Depends

Columns to be filled: AUROC, F1, Train time (populated after experiments).

8. Evaluation Metrics

8.1 Primary Metrics

Metric	Why
AUROC	Standard for binary classification; threshold-independent; comparable across models
AUPRC	More informative than AUROC when classes are imbalanced
F1 (macro)	Balances precision and recall

8.2 Secondary Metrics

Metric	Why
Accuracy	Simple interpretability for non-specialists
Precision at 90% recall	Practical: how many predictions are correct when we catch most positives?
Recall at 90% precision	Practical: how many positives do we catch when we're mostly correct?

8.3 Interpretability Metrics

Metric	How measured
Known-motif recovery rate	% of top-20 important features matching JASPAR/HOCOMOCO motifs
Cross-model feature agreement	Jaccard similarity of top-20 features between model families
Cell-type specificity of features	% of top features that differ between cell-type-specific models

9. Validity Threats

Threat	Type	Mitigation
Data leakage from overlapping regions	Internal	Chromosome-holdout split; verify no region overlap
Label noise in ENCODE annotations	External	Use high-confidence cCREs only; sensitivity analysis with stricter filters
Negative set bias	Construct	GC-matched negatives; exclude all known annotations; sensitivity analysis with multiple negative strategies
Overfitting to specific cell types	External	Cross-cell-type experiments; report per-cell-type and pooled results

Confounding by GC content	Construct	Include GC as feature; ablation experiment removing GC; GC-matched negatives
Interpretability confirmation bias	Researcher	Blind motif evaluation; compare against random features; report negative results
Sequence length artifacts	Construct	Fixed-length regions (1 kb); verify length has no predictive power
Publication bias in motif databases	External	Acknowledge that motif databases are incomplete; report novel patterns cautiously

10. Limitations

1. **Computational:** Results are limited by what Kaggle-scale compute can explore. Larger models and longer sequences may yield better accuracy.
2. **Label quality:** ENCODE annotations are computational predictions themselves (for cCREs), not all experimentally validated. Predictions are predictions of predictions.
3. **Sequence-only:** Initial phases ignore chromatin context, 3D genome structure, epigenetic marks, and expression data. Sequence alone does not capture the full regulatory picture.
4. **Cell-type coverage:** Initial experiments cover 2–3 well-annotated cell types. Results may not generalize to under-studied cell types.
5. **Causality:** The project identifies correlative patterns, not causal mechanisms. "Important features" are statistically associated with regulatory labels, not proven to cause regulatory activity.
6. **Species specificity:** All initial work is human (hg38). Cross-species generalization is not tested.
7. **Static labels:** ENCODE annotations represent steady-state regulatory landscapes. Dynamic regulation (developmental, stimulus-response) is not captured.

11. Future Publication Potential

11.1 Candidate Venue Types

Venue type	Examples	Fit
Bioinformatics workshop	MLCB (ML in Computational Biology) @ NeurIPS, ISMB workshops	Good fit for initial results
Computational biology journal	Bioinformatics, BMC Bioinformatics, PLoS Computational Biology	Good fit for full study
ML conference (applications track)	ICML, NeurIPS (application workshops)	If novel methodology emerges
Preprint	bioRxiv, arXiv (q-bio)	For early dissemination

11.2 Minimum Publication-Ready Package

1. Clear problem statement and biological motivation.
2. Reproducible data processing pipeline.
3. Multi-model comparison with standardized metrics.
4. Interpretability analysis with biological validation (motif overlap).

5. Cross-cell-type analysis.
6. Honest limitations section.
7. Public code repository.

11.3 Potential Paper Structure

1. Introduction: Non-coding regulation, interpretability gap.
2. Related work: Sequence-based regulatory prediction, interpretable ML in genomics.
3. Methods: Data, features, models, interpretability, evaluation.
4. Results: Classification performance, feature importance, motif analysis, cell-type comparison.
5. Discussion: Insights, limitations, future work.
6. Conclusion.

End of Document 04 — Research Design Document

Next: [05 — Dataset Strategy](#)